ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

# MATHEMATICS

## PAPER 1 · PRACTICE B · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central Grade 7 Mathematics paper 1 · practice b.**

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education marking guideline.

**How to mark:** each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct answers.

## QUESTION 1

**1** 1.1 D ✓ 1.2 B ✓ 1.3 C ✓ 1.4 A ✓ 1.5 B ✓ 1.6 D ✓ 1.7 A ✓ 1.8 C ✓ 1.9 C ✓ 1.10 A ✓ [10]

## QUESTION 2

**2.1** $8 \times (100 - 1) = 800 - 8$ ✓ $= 792$ ✓ (2)

**2.2** $5 + 3 = 8$ parts; $R720 \div 8 = R90$ per part ✓ (3)
Lerato: $5 \times R90 = R450$ ✓ Sipho: $3 \times R90 = R270$ ✓

**2.3** $240 \div 3$ ✓ $= 80$ km/h ✓ (2)

**2.4** (a) $R490 - R350 = R140$ ✓ (3)
(b) $^{140}/_{350} \times 100$ ✓ $= 40\%$ ✓

**2.5** $VAT = 15\% \times R800 = R120$ ✓ Price $= R920$ ✓ (2)

**2.6** Interest $= R2\,000 \times {}^{8}/_{100} \times 3$ ✓ $= R480$ ✓ Total $= R2\,480$ ✓ (3)

**[15]**

## QUESTION 3

**3.1** (a) 32 ✓ (b) $5^2 - 9 = 25 - 9$ ✓ $= 16$ ✓ (c) 4 000 ✓ (4)

**3.2** $3^4 = 81$ and $4^3 = 64$ ✓, so $3^4$ is larger ✓ (2)

**3.3** $360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5$ ✓ $= 2^3 \times 3^2 \times 5$ ✓ (2)

**[8]**

## QUESTION 4

**4.1.1** ${}^{5}/_{6} - {}^{2}/_{6}$ ✓ $= {}^{3}/_{6} = {}^{1}/_{2}$ ✓ (2)

**4.1.2** ${}^{12}/_{5} \times {}^{5}/_{4}$ ✓ $= {}^{60}/_{20}$ ✓ $= 3$ ✓ (3)

**4.1.3** ${}^{5}/_{10} + {}^{4}/_{10} + {}^{3}/_{10}$ ✓ $= {}^{12}/_{10} = 1\,{}^{1}/_{5}$ ✓ (2)

**4.2** ${}^{125}/_{1000}$ ✓ $= {}^{1}/_{8}$ ✓ (2)

**4.3** (a) 3,75 ✓ (b) $9{,}36 \div 4 = 2{,}34$ ✓✓ (3)

**4.4** ${}^{3}/_{8} \times 32$ ✓ $= 12$ learners ✓ (2)

**[14]**

## QUESTION 5

**5.1** (a) $-9 < -4$ ✓ (b) $-1 > -10$ ✓ (2)

**5.2** (a) $-14$ ✓ (b) $-7$ ✓ (c) $10$ ✓ (3)

**5.3** $-18 + 7 = -11$ (11 m below sea level) ✓ (1)

**[6]**

## QUESTION 6

**6.1.1** $18$ ✓ ; $22$ ✓ (2)

**6.1.2** $T_n = 4n - 2$ ✓✓ (2)

**6.1.3** $4(50) - 2 = 198$ ✓ (1)

**6.2** $48$ ✓; multiply each term by $2$ ✓ (2)

**6.3.1** $y = 4x + 3$ ✓✓ (2)

**6.3.2** $y = 4(10) + 3 = 43$ ✓ (1)

**6.4** $5x + 4 = 39$ ✓ $\rightarrow 5x = 35$ ✓ $\rightarrow x = 7$ ✓ (3)

**6.5** $3 \times 2^2 = 3 \times 4$ ✓ $= 12$ ✓ (2)

**[15]**

## QUESTION 7

**7.1** (a) Isosceles, obtuse-angled triangle ✓ (b) Right-angled, isosceles triangle ✓ (2)

**7.2** Any TWO: opposite sides are equal ✓; opposite sides are parallel ✓; opposite angles are equal; the diagonals bisect each other (2)

**7.3.1** Translation ✓ 6 units to the right ✓ (2)

**7.3.2** Reflection ✓ in the line *m* ✓ (2)

**7.4** (a) 5 faces ✓ (b) 5 vertices ✓ (c) 8 edges ✓ (3)

**7.5** Cube ✓ (1)

**[12]**

## QUESTION 8

**8.1** (a) 45 mm ✓ (b) 3,5 km ✓ (c) 0,75 kg ✓ (3)

**8.2.1** Large rectangle $10 \times 8 = 80$ cm$^2$ ✓ minus cut-out $4 \times 3 = 12$ cm$^2$ ✓ $\rightarrow 68$ cm$^2$ ✓ (3)
(or split into $6 \times 8 = 48$ and $4 \times 5 = 20 \rightarrow 68$ cm$^2$)

**8.2.2** $6 + 3 + 4 + 5 + 10 + 8$ ✓ $= 36$ cm ✓ (2)

**8.3** (a) $6 \times 6 \times 6$ ✓ $= 216$ cm$^3$ ✓ (b) $3 \times 3 \times 3$ ✓ $= 27$ small cubes ✓ (or $216 \div 8 = 27$) (4)

**[12]**

## QUESTION 9

**9.1.1** Soccer ✓ (1)

**9.1.2** $^{10}/_{40} \times 100$ ✓ $= 25\%$ ✓ (2)

**9.1.3** $^{15}/_{40} \times 360°$ ✓ $= 135°$ ✓ (2)

**9.2** (a) HH, HT, TH, TT ✓✓ (one mark for any three correct) (b) $^{1}/_{4}$ ✓ (3)

**[8]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**